

simplifies the problem and focuses on the key states. At the same time, the computational cost of this method is comparable to that of standard linear MPC over the same prediction time domain.

Based on the predictor in (24), design the objective function (29). The simulation lasts 300 s and the prediction step N_t is set to 20. Since we are concerned with the state of motion in the vertical direction, we set $Q = \begin{bmatrix} 10 & 0 \\ 0 & 50 \end{bmatrix}$, $R = 0.1$. As shown in Fig. 5, the simulation results demonstrate that the BE-MPC algorithm successfully achieves precise control of the DSRV at the target depth of $z = 50$ m. The figure illustrates the dynamic changes in the submersible's diving depth, vertical velocity, and control inputs (rudder angle) throughout the simulation. The initial state was selected as $x_0 = [0.2; 0; 0; 0.1; 0]^T$. The control input (rudder angle) was constrained within the range of $[-30, 30]$. From the figure, it can be observed that the controller effectively adjusts the rudder angle to optimize the motion trajectory and attitude control. During the entire simulation process, the vertical position of the submersible steadily converges to the target depth of 50 meters. This verifies that the BE-MPC algorithm can successfully perform precise control tasks in a complex nonlinear dynamic system.

V. CONCLUSIONS

In this paper, we propose a BLS-EDMD method for approximating the Koopman operator and use this method to design a new MPC. This method solves the problem of model prediction error caused by the randomness associated with the choice of basis functions in the traditional EDMD method. By leveraging the feature and enhancement layers of the BLS network, the BLS-EDMD method refines the generation of basis functions, thereby enhancing the system state representation and boosting the model's overall prediction accuracy. In simulation experiments, we apply the proposed method to the target tracking control task of a classical van der Pol oscillator system and a DSRV. Among the prediction experiments, the results show that the BLS-EDMD-based Koopman predictor has significant advantages in terms of accuracy and stability. In addition, the control accuracy of the BLS-EDMD-based MPC controller is effectively validated in a complex DSRV scenario, highlighting its potential to effectively handle high-dimensional nonlinear systems.

In future work we will further extend the current BE-MPC methodology for applications in more complex nonlinear dynamic systems, such as multi-degree-of-freedom UUV and underwater devices with more complex hydrodynamic properties.

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